

Criterion: Testing & Quality Assurance

Architecture Decision Record

Status

Status: Accepted

Date: 2026-01-05

Context

The project needs basic automated tests to reduce regressions while iterating quickly across a full-stack codebase (frontend + backend). Tests must be lightweight and runnable locally.

Decision

Use Jest for unit tests in both backend and frontend, with focused tests around error handling/config/utilities. Use Playwright for end-to-end smoke coverage in the frontend where applicable.

Alternatives Considered

Alternative	Pros	Cons	Why Not Chosen
No tests	Fastest to write features	Regressions likely	Not suitable for thesis-quality delivery
Only E2E tests	Real user flows	Slower, brittle	Too heavy for quick iteration
Only unit tests	Fast, stable	Miss integration issues	Add E2E later for key flows

Consequences

Positive: faster refactors, confidence in core helpers.

Negative: some UI routes are not fully wired, limiting E2E value.

Neutral: test coverage can be expanded when routes stabilize.

Implementation Details

Key Implementation Decisions

- * Keep unit tests small and deterministic (no external network calls).
- * Prefer testing error factories/config parsing and domain services.

Project Structure

backend/tests/
frontend/tests/

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Backend unit tests exist	[OK]	backend/tests/*
2	Frontend unit test setup exists	[OK]	Jest/RTL config + setup
3	E2E test harness exists	[OK]	Playwright config + tests folder
4	CI-friendly commands available	[!]	Depends on project scripts

Known Limitations

Limitation	Impact	Potential Solution
Limited coverage of user stories	Regressions possible in unwired routes	Add tests as routes/features are comp...